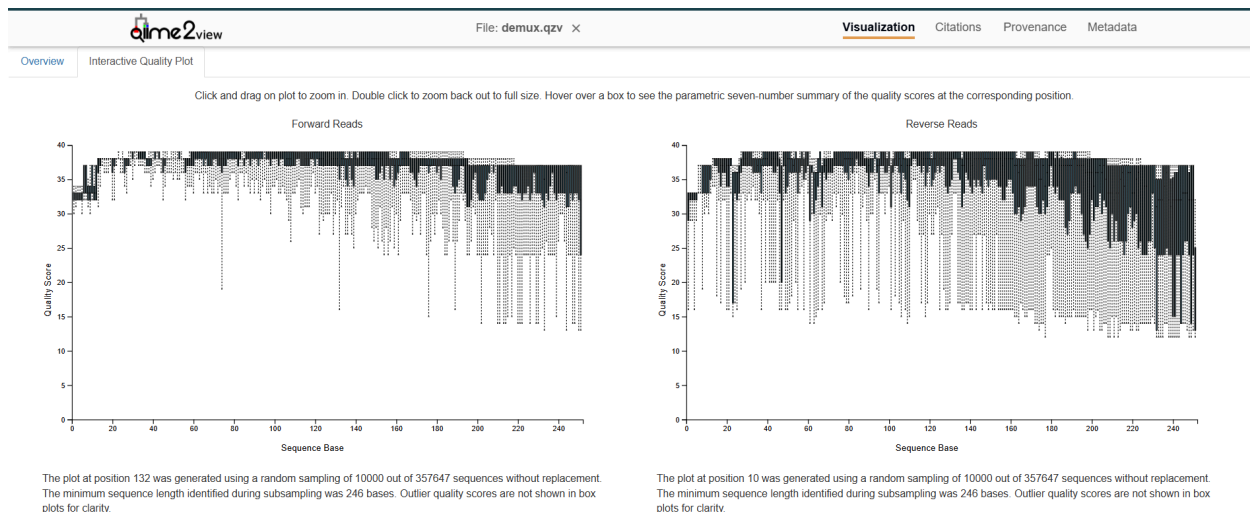


1) Examine the metadata file. What columns do you see that you think might be useful for later when doing alpha and beta diversity metrics? HINT: there are 3

The column denoting the resident populations of males and females may be useful in providing data on alpha diversity. The time point metric can be useful in determining the change in populations of species, denoting different immigration-emigration factors across species. The flock metric can be useful

2) Include a screenshot of your interactive quality plot. Based on this plot, what values would you choose for `--p-trunc-len` and `--p-trim-left` for both the forward and reverse reads? Why have you chosen those numbers? HINT: If you trim and truncate too much, you will lose too many of your reads, making your downstream analysis not useful. Think about it scientifically and only trim and truncate where the overall quality averages below 25-30.



```
qiime dada2 denoise-paired \
```

```
--i-demultiplexed-seqs demux.qza \
```

```
--p-trunc-len-f 200 \
```

```
--p-trim-left-f 0 \
```

```
--p-trunc-len-r 230 \
```

```
--p-trim-left-r 0 \
```

```
--o-representative-sequences rep-seqs.qza \
```

```
--o-table table.qza \
```

```
--o-denoising-stats stats.qza
```

In terms of the values for both graphs, I chose not to trim either sets of data as there is a suitable level of quality of reads (over or at 30) for both sets of data. In terms of the forward reads in truncating the data, I chose to cut the reads from 200 on, as it is the point where quality dips the most at a most consistent rate, while making sure not to cut more data so that there isn't enough by the end of the read. The truncating of the reverse data was a bit more challenging, as the quality of the reads were low throughout the graph, so I chose to truncate the data where the quality dips even in comparison to the average data.

3) Include a screenshot of the table summary from visualizing your table and a screenshot of the sequence length statistics from the rep-seqs file. Remember, we may eventually want to cut any samples with less than 10,000 reads. Do you see any in the interactive sample detail that might need to be cut? If so, which ones?

qiime2view

File: table.qzv ×

Overview

Interactive Sample Detail

Feature Detail

Table summary

Summary Statistic	Value
Number of samples	24
Number of features	2,141
Total frequency	293,009

QIIME 2

qiime2view

File: rep-seqs.qzv ×

Visualization

Citations

Provenance

Metadata

Sequence Length Statistics

Download sequence-length statistics as a TSV

Sequence Count	Min Length	Max Length	Mean Length	Range	Standard Deviation
2141	230	365	251.51	135	9.03

Seven-Number Summary of Sequence Lengths

Download seven-number summary as a TSV

Percentile:	2%	9%	25%	50%	75%	91%	98%
Length* (nts)	230	244	253	253	253	253	254

*Values rounded down to nearest whole number.

57_S153_L001

7,812

208_S177_L001

7,250

331_S131_L001	7,079
368_S129_L001	6,126
94_S278_L001	6,120
90_S107_L001	5,522
210_S336_L001	4,508
260_S178_L001	2,525
306_S120_L001	2,370
133_S265_L001	1,506
122_S207_L001	1,116
332_S105_L001	865
119_S106_L001	713
252_S179_L001	568

4) Once you have generated your taxonomy visualization, sort it by confidence. What are your top hits? What about if you sort by taxon? What hits do you see?

Confidence:

QIIME 2			Library	View	Forum
			File: taxonomy.qzv	Visualization	Citations Provenance Metadata
			Search:		
Feature ID # of types	Taxon category	Confidence category			
73852554604dde60b2fd7e0cb3716d5e	k__Bacteria; p__Proteobacteria; c__Gammaproteobacteria; o__Legionellales; f__Coxiellaceae; g__Rickettsiella; s__	0.9999999339764707			
08bdc6c81f6b9daab3d132b54e1d45bc	k__Bacteria; p__Actinobacteria; c__Actinobacteria; o__Actinomycetales; f__Nocardaceae; g__Rhodococcus; s__	0.9999999043577265			
7e2b74ce30f1dc7362ee86652c2118d1	k__Bacteria; p__Firmicutes; c__Bacilli; o__Bacillales; f__Paenibacillaceae; g__Saccharibacillus; s__kuersensis	0.9999998640311624			
5388d29a60ce31232943aa032947c09e	k__Bacteria; p__Chloroflexi; c__Ktedonobacteria; o__IG30-KF-AS9; f__g__; s__	0.9999998587573018			

Taxon:

QIIME 2			Library	View	Forum
			File: taxonomy.qzv	Visualization	Citations Provenance Metadata
			Search:		
Feature ID # of types	Taxon category	Confidence category			
7ed339d22e487444f24830802ada7932	k__Bacteria	0.9999949799188206			
5f5564e113fa63e2cf469cd7eb74b777	k__Bacteria	0.9999909971390686			
8aa37429cdc95e04c6241fd2b9162d99	k__Bacteria	0.9999906797657725			
14bd7a62e009f7992904b0de6f3e8f4d	k__Bacteria	0.9999889687609459			

5) When you visualize level 3 of taxonomy, what level is this? Do you see any trends as you sort by various metadata categories?

Level 3 taxonomy shows more clear distinctions in relative frequencies of bacteria across samples. When sorting the frequencies by flock, there is the most amount of change between the original visualization of the data.

6) After visualizing your filtered-table.qzv, what cutoff value will you use for generating alpha and beta diversity? Why? Include a screenshot of the interactive sample view to help justify your reasoning.

The cutoff value that I will be using will be at a 936 sampling depth, being that the value cuts a good amount of low quality reads, while maintaining an even sample size. The depth does skew the data more towards resident species over migratory, but the overall

quantity of the samples should still be enough to make an accurate display of alpha and beta diversity values.

7) The first metric we will analyze is alpha diversity. In your own words, what is alpha diversity and what are the differences between the two types of alpha diversity we will analyze (Shannon and Observed features)?

Alpha diversity has to do with diversity metrics within one specific ecosystem, encapsulating all of the metrics for species richness within the one area. In terms of Shannon vs. Observed features for looking at which types of diversity are measured, Shannon diversity would tackle both the number of each species and the ratios of species numbers between each other, while observed features would only handle the number of species present in the ecosystem.

8) Since you are looking at two metrics of diversity for 3 metadata categories, it would be helpful to make a table of the significance values. Are any of your comparisons significant? For one of the metadata data columns, there are actually 4 options. Include a screenshot of the pairwise comparisons for Shannon and Observed. Are there any significant comparisons? (HINT: look at the q value)

Comparisons of Significance Factors (p-value):

	Population	Sex	Flock
Shannon	0.457391	0.369108	0.679600
Observed	0.264548	0.413806	0.319692

In viewing all of the significance factors, The only factor that is not statistically significant would be the flock category for the Shannon significance diversity metric.

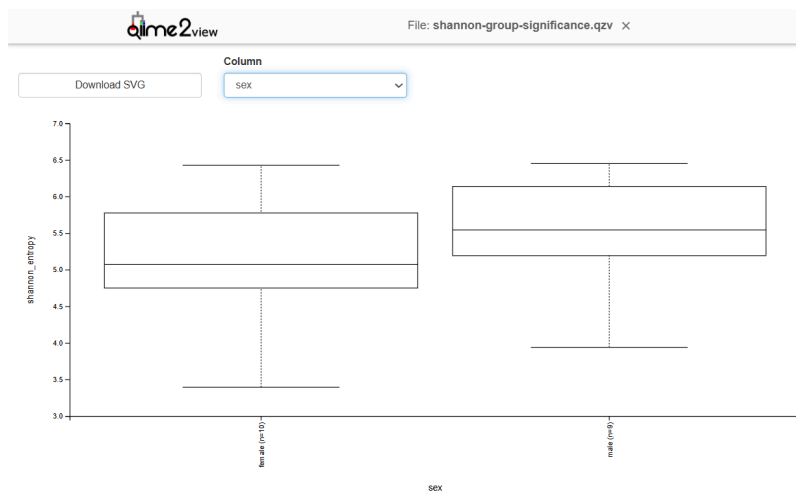
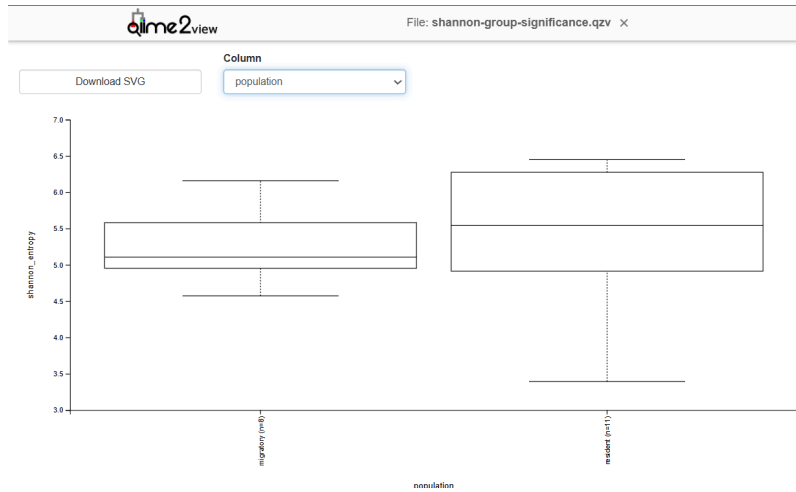
Comparison of Pairwise Significance Factors (q-value):

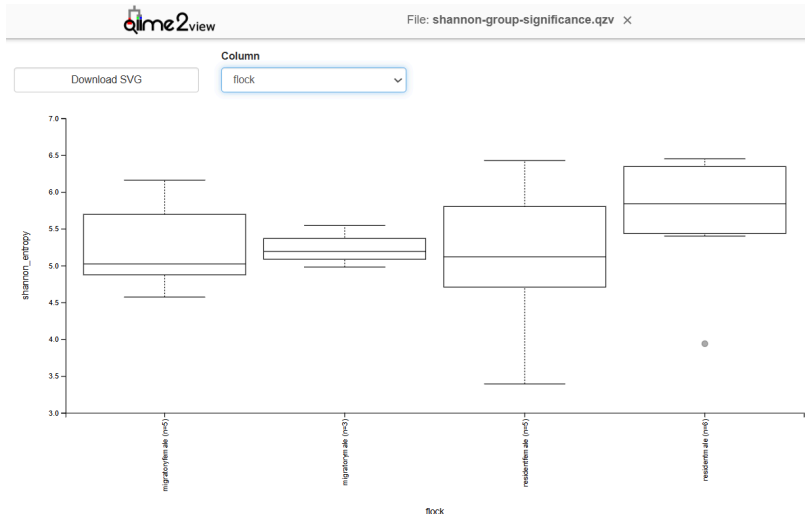
	Population	Sex	Flock
Shannon	0.457391	0.369108	0.916815 / 0.722621
Observed	0.264548	0.413806	0.754023 / 0.468534 / 0.684085

Across both shannon and observed pairwise significance factors for population and sex categories, there was statistical significance (q- values < 0.5). The flock category

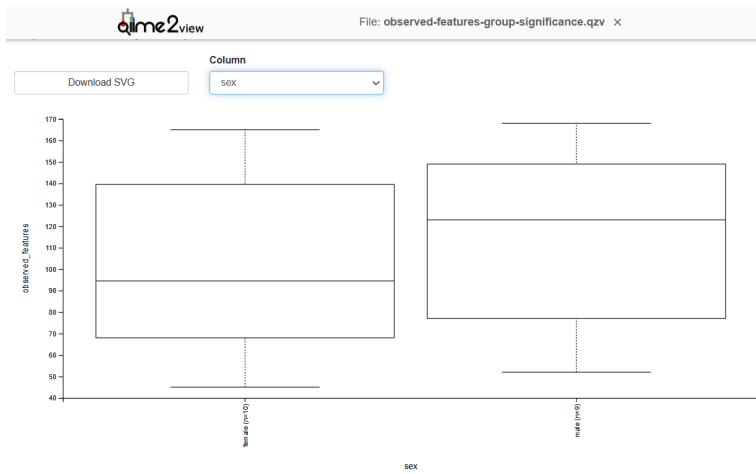
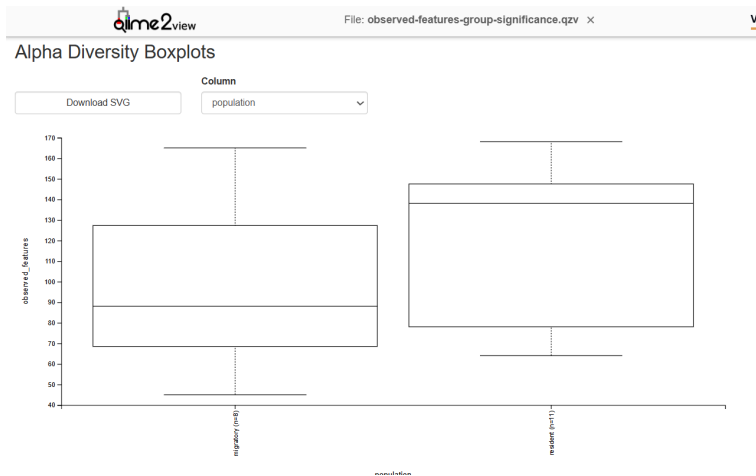
fluctuated in its significance within the samples of resident species vs. migratory, as well as sex of each species.

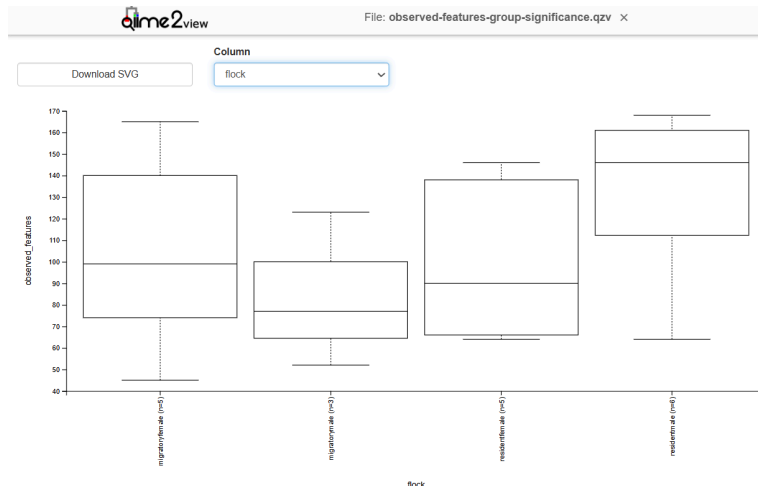
Shannon Screenshots:





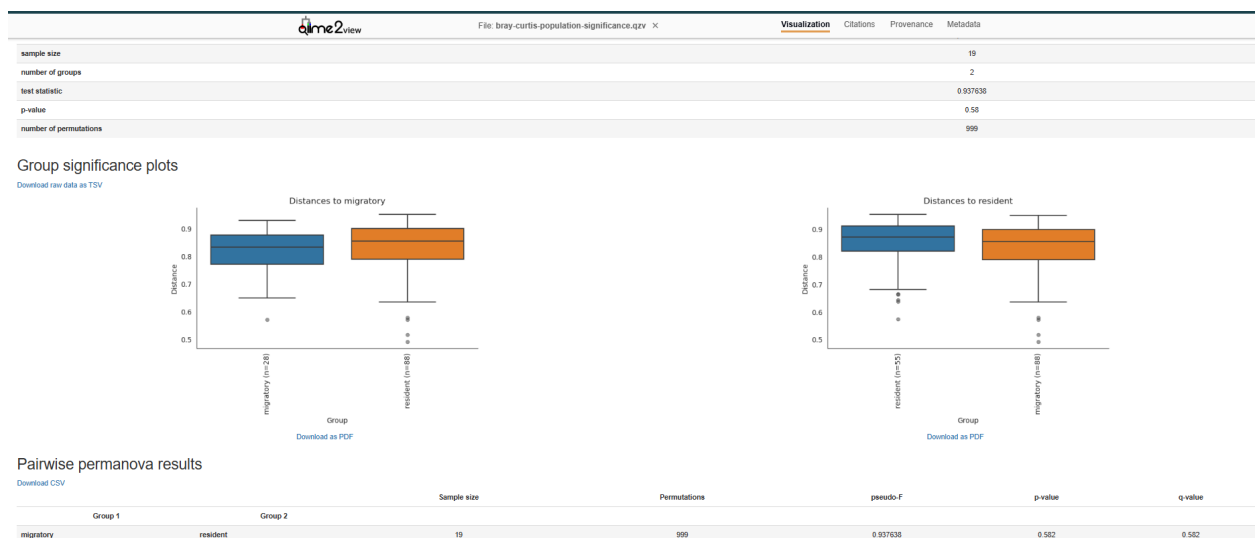
Observed Features Screenshots:





10) For beta diversity, you will need to create visualizations for Bray Curtis dissimilarity and Weighted Unifrac distance. This will require you to modify the `beta-group-significance` code. You should have one visualization for each metadata column you are interested in. Include a screenshot of each visualization. Is there any significance? Regardless of significance, how can you interpret these results (hint: what is beta diversity looking at?) How are Bray Curtis dissimilarity and Weighted Unifrac distance different?

Bray Curtis Population:

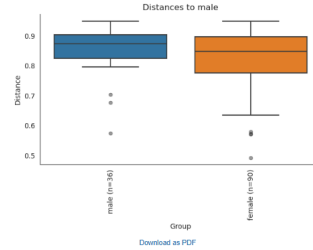
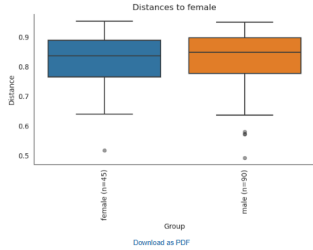


Bray Curtis Sex:

	qime2view	File: bray-curtis-sex-significance.qzv	Visualization	Citations	Provenance	Metadata
sample size						19
number of groups						2
test statistic						0.717385
p-value						0.959
number of permutations						999

Group significance plots

[Download raw data as TSV](#)



Pairwise permanova results

[Download CSV](#)

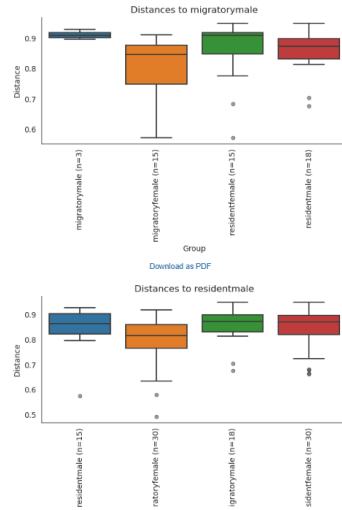
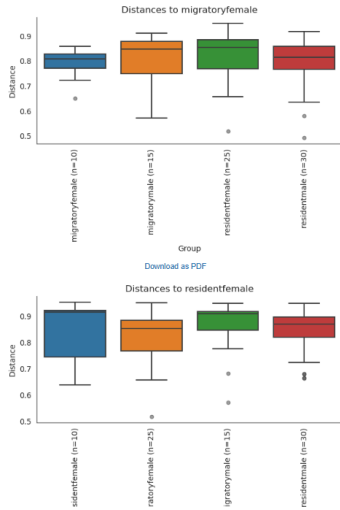
Group 1	Group 2	Sample size	Permutations	pseudo-F	p-value	q-value
female	male	19	999	0.717385	0.953	0.953

Bray Curtis Flock:

	qime2view	File: bray-curtis-flock-significance.qzv	Visualization	Citations	Provenance	Metadata
--	-----------	--	---------------	-----------	------------	----------

Group significance plots

[Download raw data as TSV](#)



	qime2view	File: bray-curtis-flock-significance.qzv	Visualization	Citations	Provenance	Metadata
--	-----------	--	---------------	-----------	------------	----------

Overview

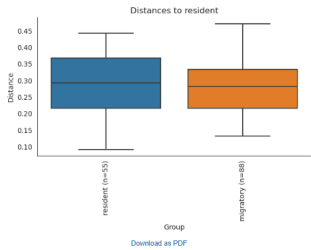
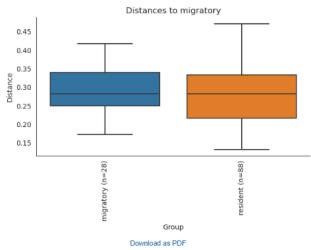
method name	PERMANOVA results
test statistic name	PERMANOVA
sample size	19
number of groups	4
test statistic	0.854771
p-value	0.878
number of permutations	999

Weighted Unifrac Population:

		File: weighted-unifrac-population-significance.qzv X	Visualization	Citations	Provenance	Metadata
sample size	19					
number of groups	2					
test statistic	0.596992					
p-value	0.662					
number of permutations	999					

Group significance plots

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Pairwise permanova results

[Download CSV](#)

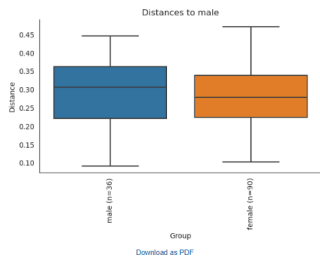
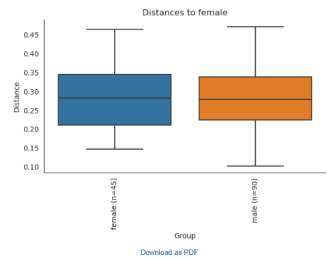
Group 1	Group 2	Sample size	Permutations	pseudo-F	p-value	q-value
migratory	resident	19	999	0.596992	0.662	0.662

Weighted Unifrac Sex:

		File: weighted-unifrac-sex-significance.qzv X	Visualization	Citations	Provenance	Metadata
sample size	19					
number of groups	2					
test statistic	0.426128					
p-value	0.855					
number of permutations	999					

Group significance plots

[Download raw data as TSV](#)

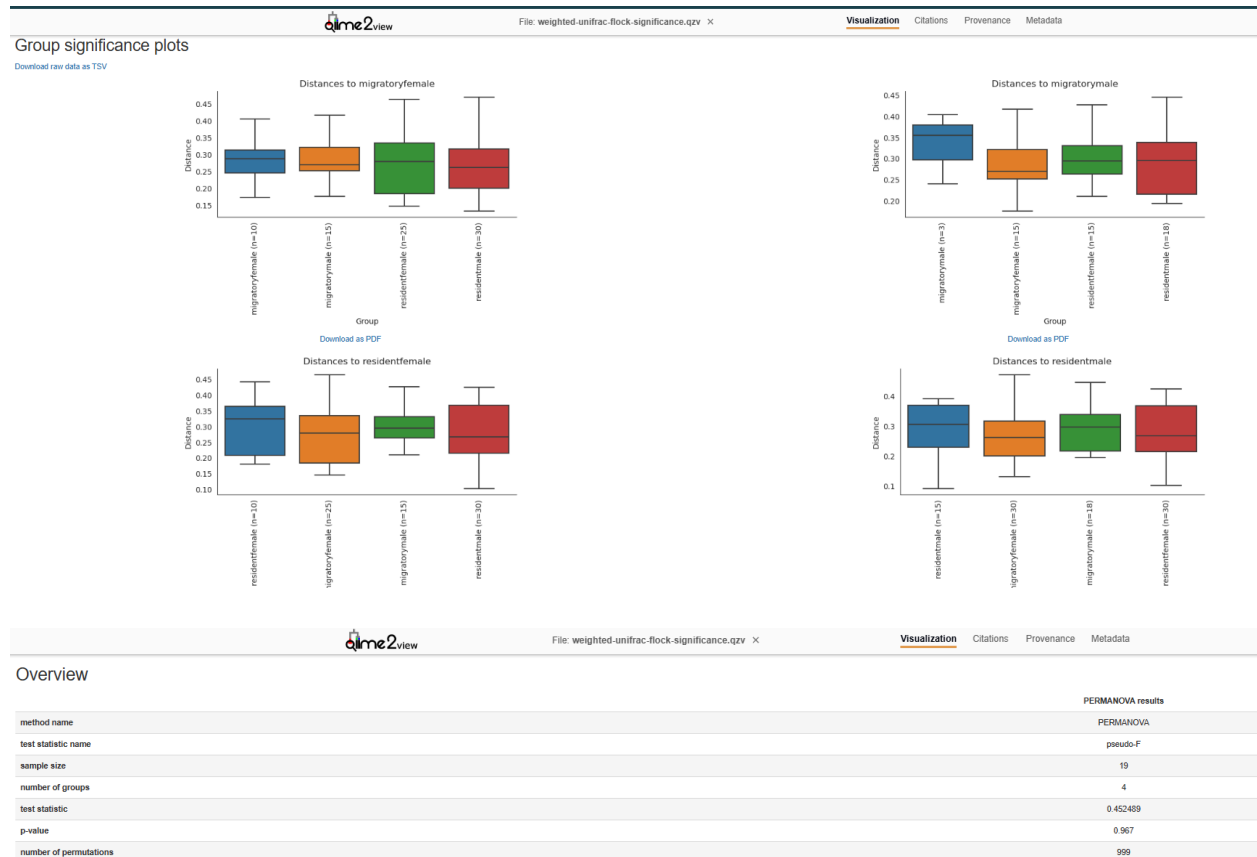


Pairwise permanova results

[Download CSV](#)

Group 1	Group 2	Sample size	Permutations	pseudo-F	p-value	q-value
female	male	19	999	0.426128	0.872	0.872

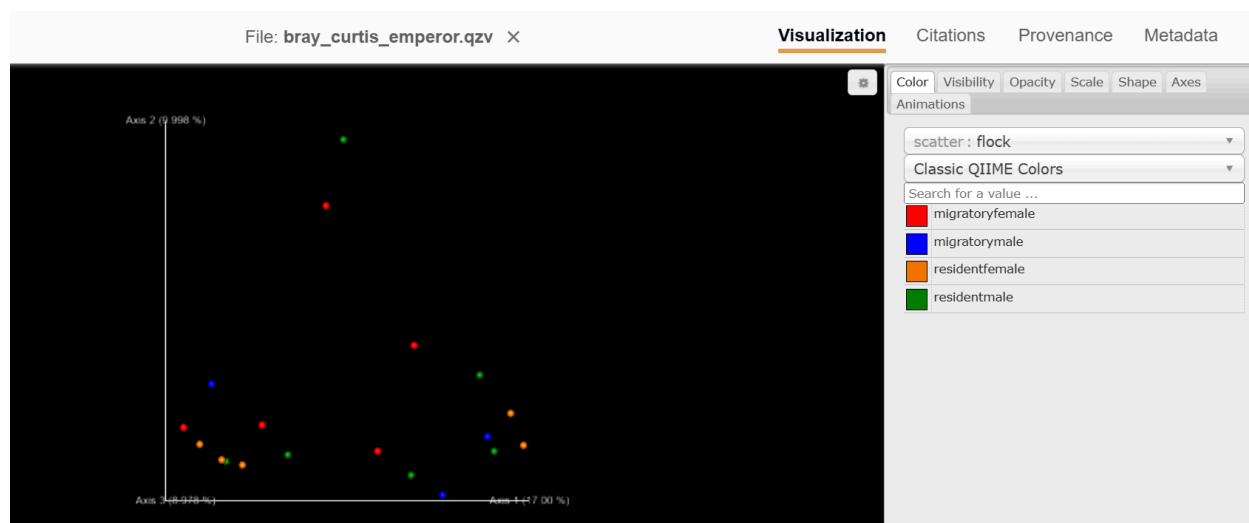
Weighted Unifrac Flock:



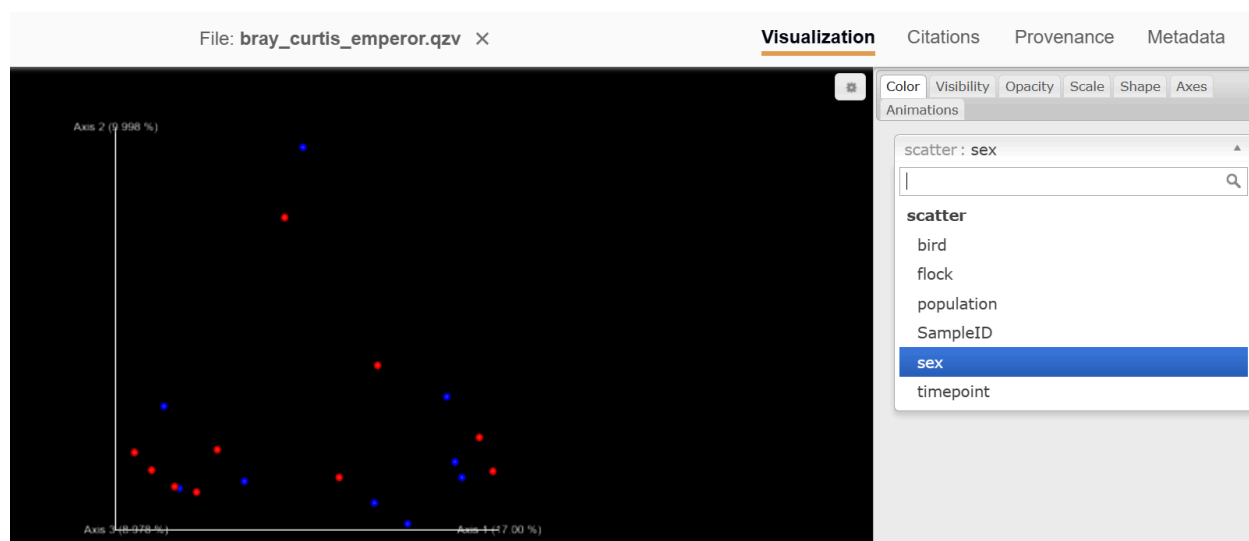
We can interpret the differences in Bray-Curtis and Weighted Unifrac analysis based on the distance of the data between the samples in each visualization. The Bray-Curtis distances being further apart than the Weighted Unifrac distances underlines the defining characteristics of their focus, being that the Bray-Curtis does not take phylogenetic similarities into account for comparisons, while Unifrac does, aligning each subset of data closer together based on similarities in that regard.

11) The `core-metrics-phylogeny` command generates a file called `bray-curtis-emporer.qzv`. Include 3 screenshots total (where the points are colored based on the metadata metrics). How do these results help you make sense of the results you got from question 10?

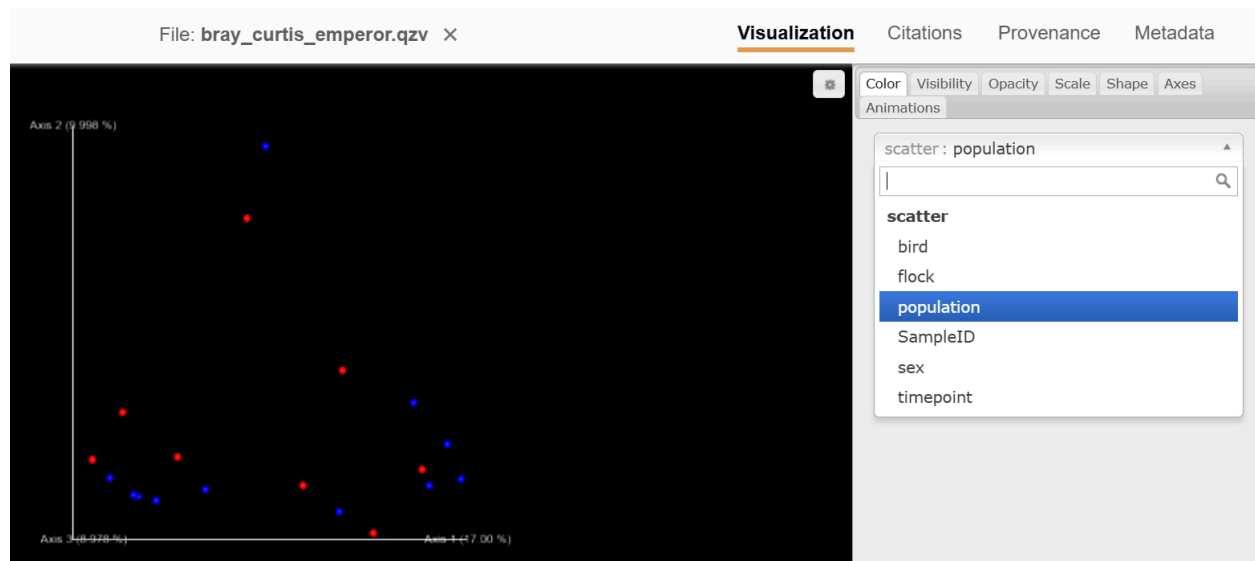
Flock:



Sex:

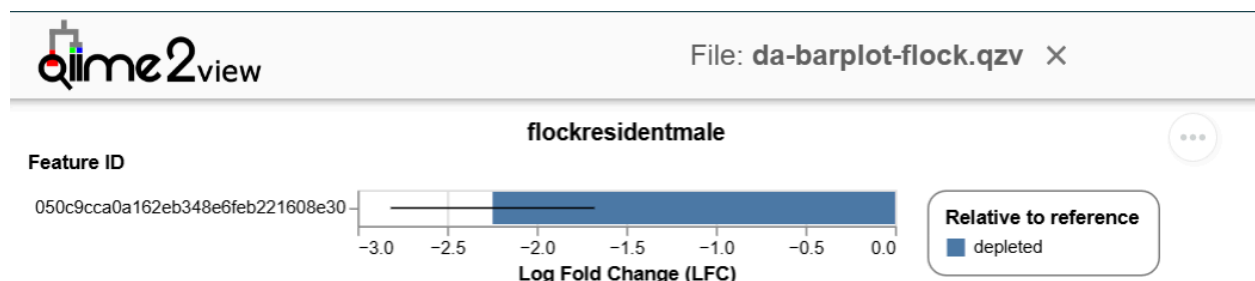


Population:

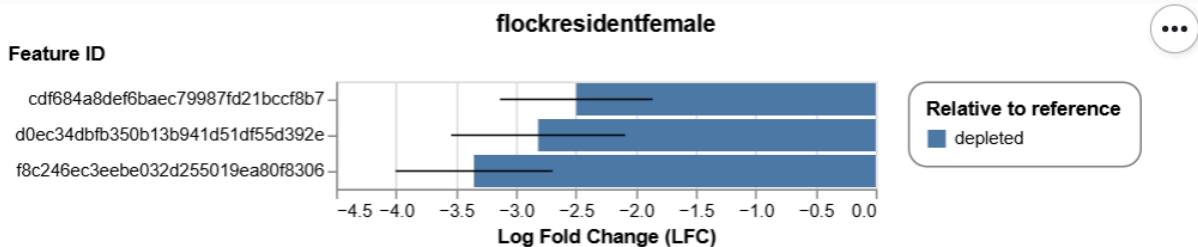


All three of the graphs showed similar types of groupings based on their respective characteristics, all also showing a clustering of data across the graph, highlighting a low level of statistical significance of data across all metrics.

12) Using ANCOMBC, do you find any specific taxa that are differentially expressed with your three metadata categories? Please include a screenshot of any differentially expressed taxa and identify the species using the taxonomy.qzv file.



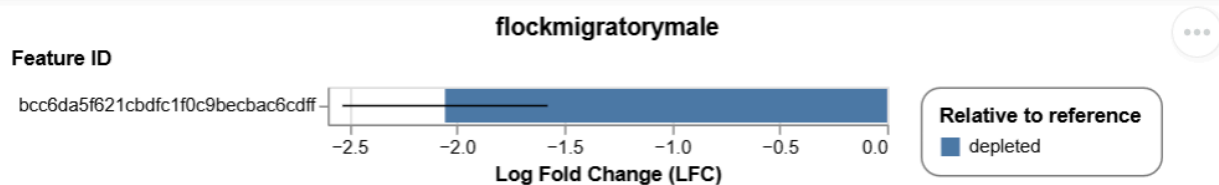
k__Bacteria; p__Acidobacteria; c__Acidobacteriia; o__Acidobacteriales; f__Acidobacteriaceae; g__; s__



k__Bacteria; p__Bacteroidetes; c__Flavobacteriia; o__Flavobacteriales; f__Flavobacteriaceae; g__Flavobacterium; s__

k__Bacteria; p__Proteobacteria; c__Gammaproteobacteria; o__Pseudomonadales

k__Bacteria; p__Bacteroidetes; c__Flavobacteriia; o__Flavobacteriales; f__Flavobacteriaceae



k__Bacteria; p__Actinobacteria; c__Actinobacteria; o__Actinomycetales